GG-M-105D
July 18, 1977
SUPERSEDING
Int. Fed. Spec.
GG-M-00105c(DSA-DM)
October 2, 1964 and
Fed. Spec. GG-M-105b
May 27, 1957

FEDERAL SPECIFICATION

MALLET, ORAL SURGERY

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration for the use of all Federal Agencies.

1. SCOPE

1.1 Scope. This specification covers a mallet for use in oral surgery.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123	- Marking for Shipment (Civil Agencies).
Fed. Std. No. 151	- Metal, Test Methods.

FSC 6520

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116	- Methods of Preservation-packaging.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap: Water-Vaporproof or Waterproof, Flexible.
MIL-M-20693	- Molding Plastic, Polyamide (Nylon), Rigid.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

Drawing:

Defense Personnel Support Center (DPSC)

20383	- Mallet, Oral Surgery, Hand Type, Mead.

(Copies of Military Specifications, Standards and Drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Steel. The steel shall be class 4, type 416 stainless steel in accordance with GG-I-526.

3.1.2 Plastic. Plastic shall be high strength polyamide (nylon) resin in conformance with type I of MIL-M-20693.

3.2 Style, design and dimensions. Mallets shall be of the type known to the dental profession as "Mead", and shall be suitable for use in dental and surgical procedures. Style, design, and dimensions of the mallet shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 20383.

3.2.1 Drawings. The drawings forming a part of this specification are not production drawings. Drawings establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the drawings to provide a product with proper design, configuration, fit and function. No deviations from the dimensional limitations prescribed on the drawings are permissible without prior approval of the contracting officer.

3.3 Construction.

3.3.1 Handle. It shall be a replaceable handle made of plastic material as specified in 3.1.2. The handle shall be solid and shall have a polished finish.

3.3.2 Head. The head shall be made of steel as specified in 3.1.1. It shall be threaded on both ends to accommodate the retaining rings and shall be securely attached to the handle by mechanical means, through the use of a pin. The head shall have a mirror or satin finish.

3.3.3 Retaining rings. The retaining rings shall be made of steel as specified in 3.1.1. Each ring shall be threaded to fit either threaded end of the head. The outer cylindrical surface of the ring shall be knurled.

3.3.4 Disks. Disks shall be made of plastic as specified in 3.1.2. The disks shall seat securely without wobble when the retaining rings are fully screwed on the head. Two spare disks shall be furnished with each mallet.

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3.4 Corrosion tests. The steel head and retainer rings shall meet the boil and copper sulfate test requirements of GG-I-526.

3.5 Sterilization. The plastic handle and disks shall show no visual evidence of warping, expanding, shrinking, cracking, deformation, or any other injury, when subjected to the sterilization test specified in 4.3.2.

3.6 Identification marking. Each mallet shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.7 Workmanship. Workmanship shall be first class throughout. Mallets shall be free from defects which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the metal and plastic, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the date of testing, the test method, individual test results, the specification requirements, the lot number, the lot size and the sample size.

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4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Mallets shall be examined for defects, including but not limited to those listed in table I.

TABLE I. Classification of defects

Categories	Defects
Major:	
101	Design and construction not as specified.
102	Head not securely attached to handle.
103	Handle not solid.
104	Head not threaded on both ends to accommodate the threaded retaining rings.
105	Retaining rings not threaded to fit either threaded end of head.
106	Disks do not seal securely without wobble when the knurled retaining rings are fully screwed on the head.
Minor:	
201	Handle does not have a polished finish.
202	Head does not have a mirror or satin finish.
203	Outer surface of rings not knurled.
204	Dimensions (other than 104 or 105) not as specified.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table II. The unit of product for examination shall be one mallet.

TABLE II. Examination criteria

Type of defect	Inspection level	AQL (Percent defective)
Major	S-2	1.5
Minor	S-2	4.0

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4.2.2 Testing. Mallets shall be tested as specified in table III and 4.3. Inspection levels and acceptable quality levels (AQL's) shall be as listed in table III. The unit of product for testing shall be one mallet.

TABLE III. Test criteria

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Head and retainer rings	Chemical analysis	3.1.1	4.3.1	**	*
	Boil test	3.4	GG-I-526	S-2	1.5
	Copper sulfate	3.4	GG-I-526	S-2	1.5
Handle and disks	Sterilization	3.5	4.3.2	S-2	1.5

* Acceptance number shall be zero for all sample sizes.
** A sufficient amount from each lot of metal used in the construction of the mallets shall be selected for chemical analysis.

4.2.3 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 expressed in terms of percent defective.

TABLE IV. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

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6.2 Specification changes. Because of the extensiveness of changes to

4.2.3.1 Examination of closure, waterproofing and banding of container. When the container is required to comply with PPP-B-585, PPP-B-601, PPP-B-621 or PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

4.3 Tests.

4.3.1 Chemical analysis. Metal analysis shall be conducted in accordance with method 111 or 112 of Fed. Std. No. 151.

4.3.2 Sterilization test. The finished mallets shall be steam-autoclaved under 27 pounds per square inch gauge pressure at 270° Fahrenheit (F) for 20 minutes. The assembled mallet (finished unit) shall be examined for compliance with the requirements of 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Each mallet and two spare disks shall be thoroughly cleaned with the applicable method specified in MIL-P-116 and wrapped in neutral tissue or other suitable anti-corrosive material. The mallet shall be further packaged in accordance with MIL-P-116, method IC-1 except that a preservative shall not be applied.

5.1.2 Level C. Mallets and spare disks shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2. Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Sixty (60) mallets and spare disks shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. Sixty (60) mallets and spare disks shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity and spare disks shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Each unit container and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-540-2050	MALLET, ORAL SURGERY, Hand Type, Metal

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Navy - MS
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DLA-DM

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Project No. 6520-1421

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DOCUMENT IDENTIFIER AND TITLE
GG-M-105D MALLET, ORAL SURGERY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

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2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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